

Digits, places, and hex

Name: _____ Friday, September 25. Work alone or in pairs. **Core: A-D. Extensions: E, F.**
Show the places you added, not just the answer.

bit	7	6	5	4	3	2	1	0
place is worth	128	64	32	16	8	4	2	1

CORE A. Counting with fewer symbols

- How many symbols can one base-10 digit use? One base-2 digit? One hex character? List them.
- Count from zero to eight in base 2. Which number is the first that needs a fourth bit, and why?

CORE B. Binary and decimal

Read these as decimal. Write the places you added.

- 1111 = _____
- 10101 = _____
- 1011 0011 = _____

Now go the other way: pick the places that add up to the number and write all eight bits.

4. 25 =

--	--	--	--	--	--	--	--

5. 100 =

--	--	--	--	--	--	--	--

CORE C. Build the hex chart

Fill every empty cell. You'll use this table in D.

Decimal	Binary (4 bits)	Hex
0	0000	0
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		A
11		
12		
13	1101	
14		
15		

CORE **D. Hex you have already seen**

Every color on a web page can be written as a code like #FF0000. It is three bytes in hex: two characters for **red**, two for **green**, two for **blue**. Each byte says how much of that light to use, from 0 (none) to 255 (all of it).

- 1. #FF0000: how much red, green, and blue is that, in decimal? What color is it? _____
- 2. **Mystery color:** #4EA1D6. Fill the table, then predict the color before you look it up.

	Red 4E	Green A1	Blue D6
Eight bits			
Decimal			

- My prediction: _____
- Now check it: search #4EA1D6 (most search engines show the color), or paste it into any online color picker. Were you close?
- 3. #808080 has the same amount of red, green, and blue. What do you think it looks like? What about #000000 and #FFFFFF?

- 4. **Make your own.** Pick three amounts from 0 to 255, write them as a hex color code, and give only the code to a partner. Can they tell roughly what color it is before looking it up?

EXTENSION **E. How many patterns?**

Width (bits)	2	3	4	5	6	7	8	9	10
Different patterns	4								
Largest value	3								

- 1. Write the general rule: with *n* bits, there are _____ patterns and the largest value is _____.
- 2. **Twenty questions.** A partner picks a whole number from 1 to 100. You may ask only yes/no questions. What is the fewest questions that **guarantees** you find it, no matter the number? Each answer is one bit. Play one round and count.

- 3. How many different colors can a code like #4EA1D6 describe? _____
- 4. **The kill screen.** The arcade game *Pac-Man* keeps the level number in one byte, counting from zero, so level 1 is stored as 0000 0000. What is stored at the highest level the byte can hold? Players who reach level 256 find half the maze turned to garbage. What goes wrong when the game adds one to that byte?

EXTENSION **F. Clock speed, again**

A processor runs at **2.5 GHz** (2.5 billion cycles per second).

- 1. How many cycles does it complete in one millisecond (1/1000 of a second)? _____
- 2. If an instruction takes **5 cycles** on average, how many instructions can it run in that millisecond?
